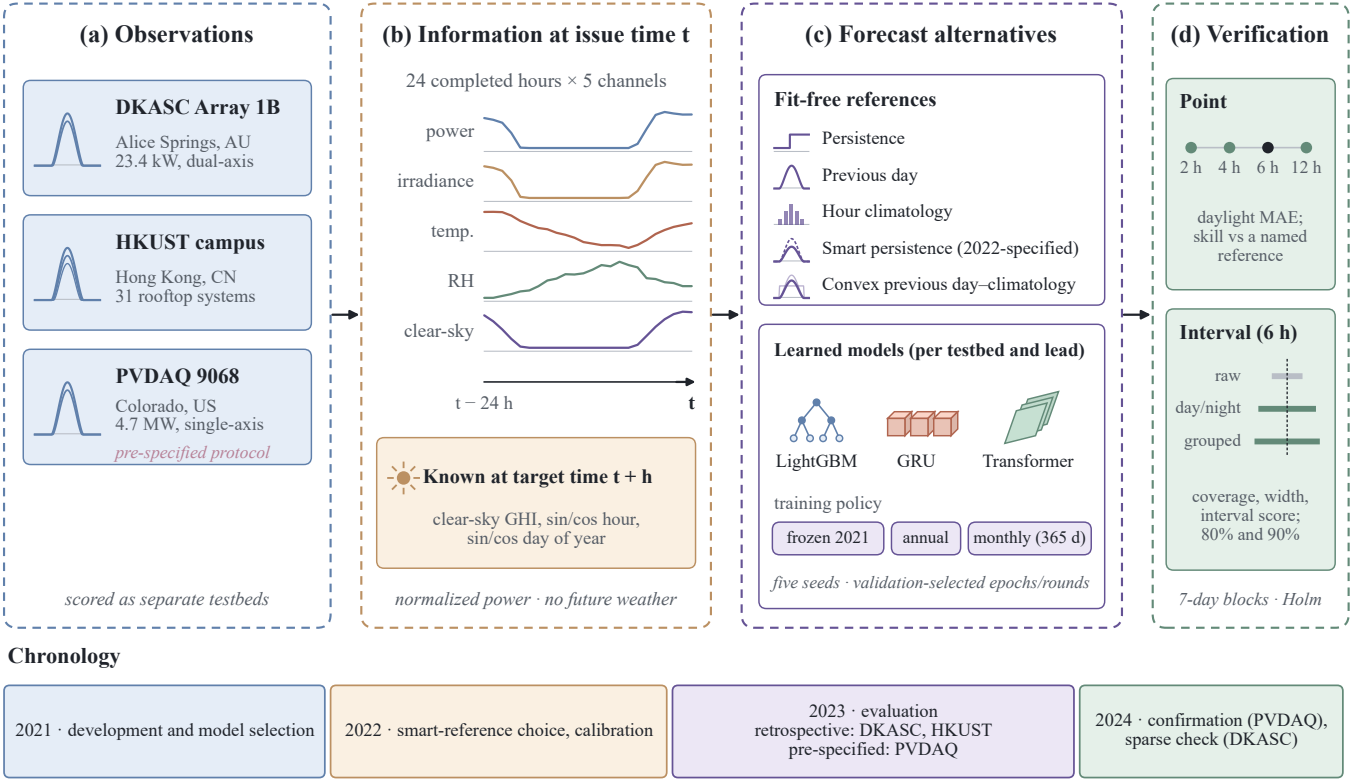

(a) Observations
DKASC Array 1B
Alice Springs, AU
23.4 kW, dual-axis
HKUST campus
Hong Kong, CN
31 rooftop systems
PVDAQ 9068
Colorado, US
4.7 MW, single-axis
pre-specified protocol
scored as separate testbeds
(b) Information at issue time t
24 completed hours × 5 channels
power
irradiance
temp.
RH
clear-sky
t − 24 h
t
Known at target time t + h
clear-sky GHI, sin/cos hour, sin/cos day of year
normalized power · no future weather
(c) Forecast alternatives
Fit-free references
Persistence
Previous day
Hour climatology
Smart persistence (2022-specified)
Convex previous day–climatology
Learned models (per testbed and lead)
LightGBM
GRU
Transformer
training policy
frozen 2021
annual
monthly (365 d)
five seeds · validation-selected epochs/rounds
(d) Verification
Point
2 h
4 h
6 h
12 h
daylight MAE; skill vs a named reference
Interval (6 h)
raw
day/night
grouped
coverage, width, interval score; 80% and 90%
7-day blocks · Holm
Chronology
2021 · development and model selection
2022 · smart-reference choice, calibration
2023 · evaluation
retrospective: DKASC, HKUST
pre-specified: PVDAQ
2024 · confirmation (PVDAQ), sparse check (DKASC)